5/4/2026

Fitness Activity Dashboard

CA 2

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Summary

This CA involved the development of an interactive fitness activity dashboard using Shiny for Python. The aim of it was to let a general audience explore daily physical activity patterns and understand how different variables (steps, distance, activity intensity, calories burned, etc.) relate to each other.

The dashboard is designed for users interested in fitness and health behaviour. It includes a landing page, overview statistics, activity trend visualisations, calorie relationship analysis and user comparison using clustering. The app allows users to filter the data by user ID, date range, and selected activity. It also includes Plotly charts, a Seaborn correlation heatmap and a k-means clustering section to group users into activity profiles.

The application helps users in identifying overall trends, comparing activity across weekdays, understanding which measures are strongly related to calories burned, and observing differences between individual users.

Background Research

Before designing the dashboard, two similar Kaggle analyses were reviewed. They helped in understanding which variables and visualisations were most useful for this analysis.

The first analysis reviewed was Fitbit Fitness Tracker Data by Nada Emad. Their notebook also explored the tracker data and focused on common activity variables, such as steps, calories, distance, and active minutes. It helped confirm that visualisations such as activity summaries, calorie comparisons, and active minute breakdowns are appropriate for this CA.

The second analysis reviewed was Fitbit Data Analysis by Pramit Khatri. This analysis also focused on a similar approach and showed how user behaviour can be explored through daily activity, steps, calories, and activity intensity. It helped me with making the decision of including charts showing activity trends, relationships with calories burned, and user-level comparisons.

Both studies showed that Fitbit-style activity data is useful when presented through simple and understandable visualisations. This influenced the structure of my own dashboard, which combines overview statistics, time-series charts, weekday comparisons, scatter plots, a correlation heatmap, and user clustering.

Data

Data Source

The original dataset can be found under the following link: Kaggle.

Data Description

The dataset contains daily activity records for 35 anonymised users. The date ranges from 12/03/2026 to 12/05/2026. After combining the two original .csv files, the final cleaned dataset contained 1373 daily records.

The main variables used in the dashboard included the following: user ID, activity date, total steps, total distance, very active minutes, fairly active minutes, lightly active minutes, sedentary minutes, and calories burned.

Several additional variables were created to support the visualisations: weekday, month, total active minutes, sedentary hours, active percentage, calories/step, calories/distance, step category, and day type. The user summary dataset was also created by grouping the data by user ID and calculating average values for each user.

Data Cleaning

First, both original .csv files were imported and combined into one data frame. The activity date was converted into a proper datetime format.

New columns were created to make the data more useful for the analysis. Total active minutes were calculated by adding all other minute columns. Sedentary hours were calculated from sedentary minutes. Weekday and month columns were also created from the activity date.

Step categories were added to classify each day. A weekday/weekend column was also created.

Some records contained zero steps/zero distance. They were kept, because they may represent inactive or rest days. However, calculations for some fields were only completed for valid values.

Criticism

Key Challenges & Technical Issues

One of the main challenges was finding an appropriate dataset. A lot of datasets out there were artificial, therefore did not support meaningful visualisation. The Fitbit activity dataset was suitable, because it contained clear numerical variables and was a real-world example.

Another challenge was deciding which features were most important for the dashboard. The decision was to choose the ones that were most important for the general audience. The focus was on total steps, total distance, calories, active minutes, sedentary minutes, and activity intensity, because these were easy to understand and linked clearly to the aim of the project.

Time management was also a challenge. This CA had to be completed alongside other coursework. It was important to keep the dashboard focused and try not to include every possible chart in just five pages. Instead, the focus was on to make it clear and readable for everyone.

One of the technical challenges was date formatting. The dates that appeared in the app as nanosecond timestamp values instead of regular dates. This was fixed by converting the datetime column correctly and creating a formatted column that could be used for display.

Another technical issue was file path handling. Initially, the app depended on absolute paths from the local computer, but it wouldn't work when deploying the app or moving the project. The issue was fixed by using relative paths based on the location of the *.py* file, so the app could load all other necessary files from the project folder.

Key Strengths

The first key strength of the dashboard is the clear multi-page structure. It makes the dashboard easy to follow and suitable for general audience.

The second key strength is the interactivity. Users can filter by user, change the date range, choose different activity metrics, toggle trendlines and adjust clustering settings. This allows them to explore the dataset themselves rather than viewing static charts.

Main Weakness

The main weakness is the limited context of the dataset. The users are anonymous and the data does not include any demographic information. So, the dashboard can show activity patterns, but it cannot fully explain why users differ from each other or make health-related conclusions.

Deployed Application

The deployed app can be found here: Fitness Activity Dashboard - Albert Skalinski CA 2.

Additional Information

The project has been developed with the aid of AI (ChatGPT). All the prompts and the entire conversation can be found here: ChatGPT.

The repo with source files can be found here: GitHub.

Version History

29/04/2026

- Chose the dataset
- Created the repo
- Established a pipeline
- Created README.md

04/05/2026

- Cleaned the dataset
- Created derived fields
- Performed EDA
- Decided on the structure of the app
- Cluster each user
- Wrote the logic for the app
- Implemented .css
- Tested the app
- Added interpretations
- Added requirements.txt
- Fixed paths
- Deployed the app
- Recorded the screencast
- Wrote the report